Nova Analytics - EDA Report

Dataset: auto-mpg.csv

398 rows x 9 columns

Dataset Information

Shape

Rows	**398**
Columns	**9**

Data Types

Column	Data Type	Non-Null	Null	Unique
mpg	float64	398	0	129
cylinders	int64	398	0	5
displacement	float64	398	0	82
horsepower	object	398	0	94
weight	int64	398	0	351
acceleration	float64	398	0	95
model year	int64	398	0	13
origin	int64	398	0	3
car name	object	398	0	305

Descriptive Statistics

Column	Mean	Median	Std Dev	Min	25%	50%	75%	Max
mpg	23.51	23.00	7.82	9.00	17.5	23.0	29.0	46.60
cylinders	5.45	4.00	1.70	3.00	4.0	4.0	8.0	8.00
displacement	193.43	148.50	104.27	68.00	104.25	148.5	262.0	455.00
weight	2970.42	2803.50	846.84	1613.00	2223.75	2803.5	3608.0	5140.00

acceleration	15.57	15.50	2.76	8.00	13.82	15.5	17.18	24.80
model year	76.01	76.00	3.70	70.00	73.0	76.0	79.0	82.00
origin	1.57	1.00	0.80	1.00	1.0	1.0	2.0	3.00

Missing Value Analysis

Total Missing Values: **0**

Column	Missing Count	Percentage
mpg	0	0.0%
cylinders	0	0.0%
displacement	0	0.0%
horsepower	0	0.0%
weight	0	0.0%
acceleration	0	0.0%
model year	0	0.0%
origin	0	0.0%
car name	0	0.0%

No missing values detected in this dataset.

Duplicate Analysis

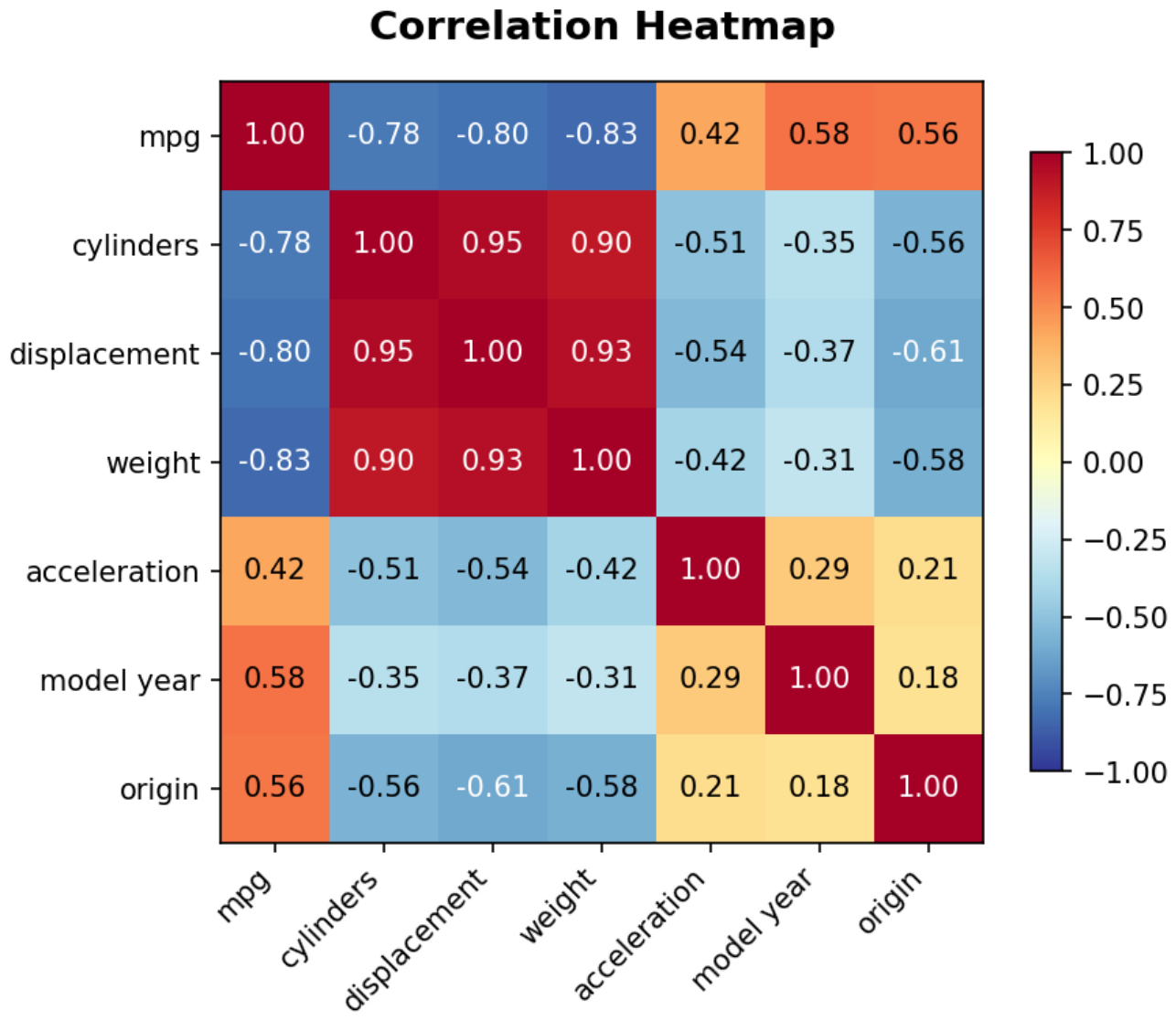
Duplicate Rows	**0 (0.0%)**
Total Rows	**398**

Correlation Analysis

Correlation Matrix

	mpg	cylinders	displacement	weight	acceleration	model year	origin
mpg	1.00	-0.78	-0.80	-0.83	0.42	0.58	0.56

cylinders	1.00	0.95	0.90	-0.51	-0.35	-0.56
displacement	0.95	1.00	0.93	-0.54	-0.37	-0.61
weight	0.90	0.93	1.00	-0.42	-0.31	-0.58
acceleration	-0.51	-0.54	-0.42	1.00	0.29	0.21
model year	-0.35	-0.37	-0.31	0.29	1.00	0.18
origin	-0.56	-0.61	-0.58	0.21	0.18	1.00

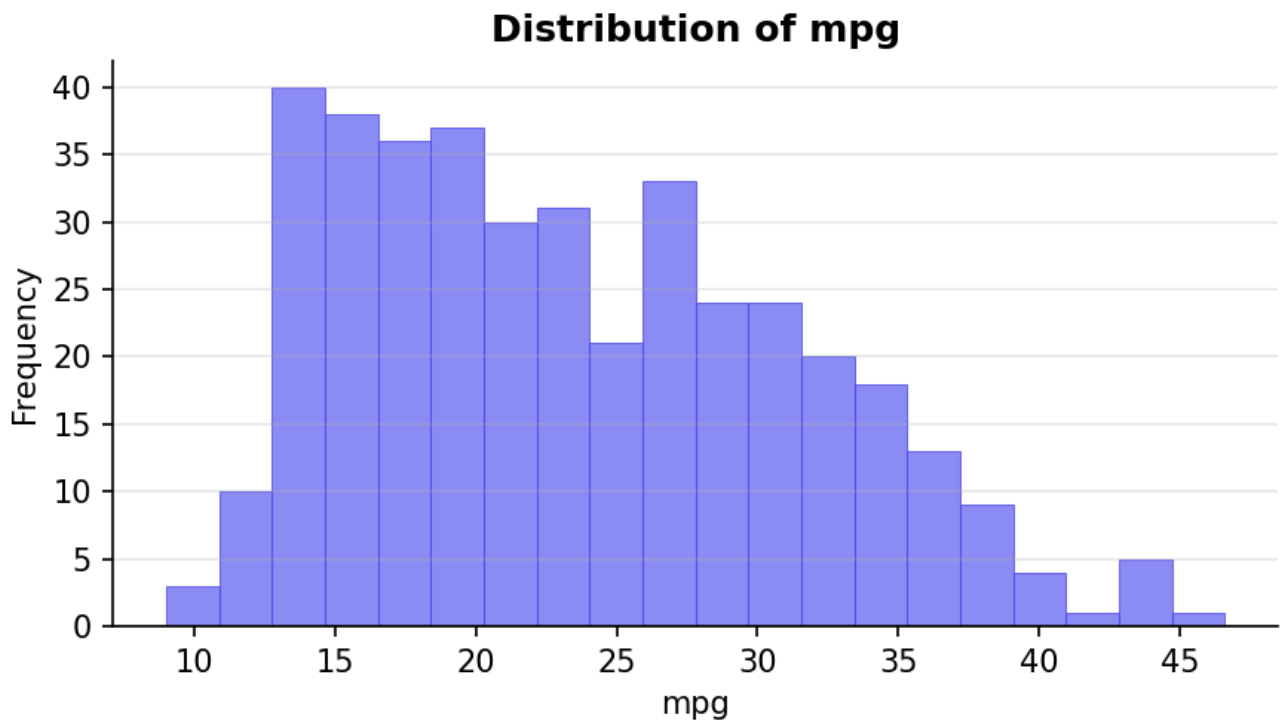


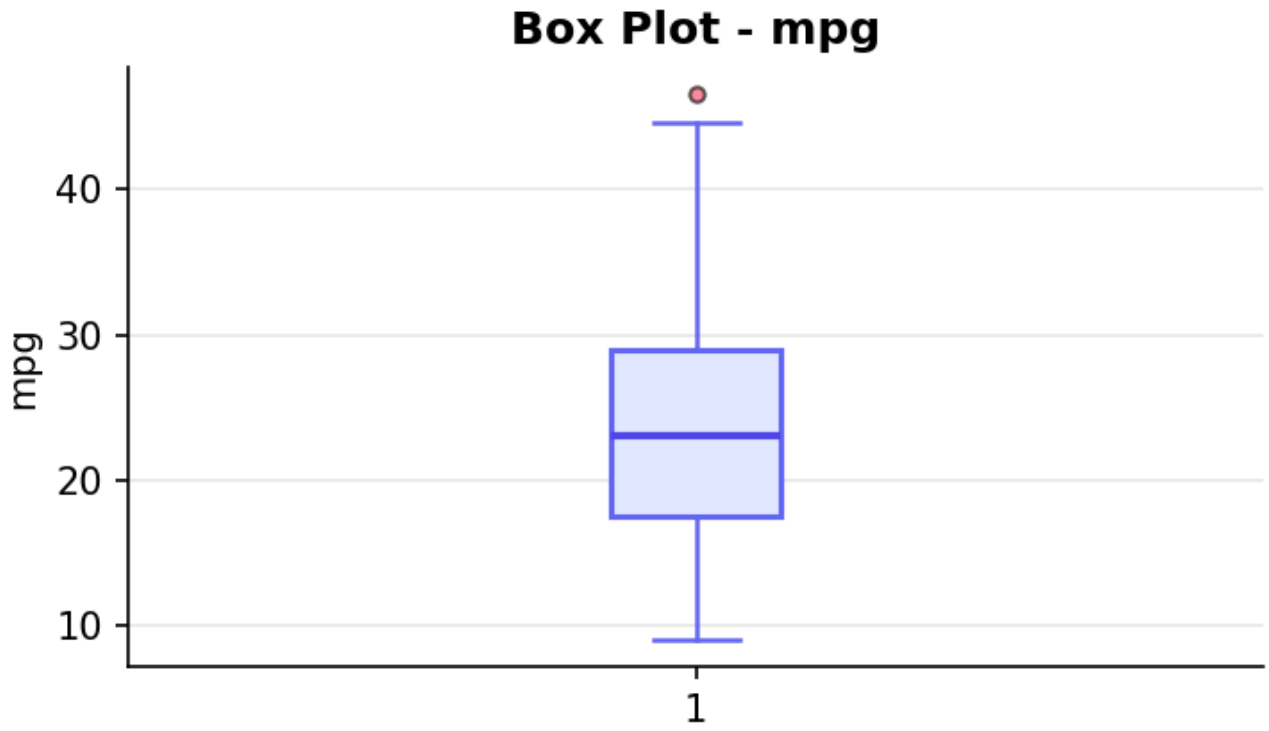
Outlier Analysis

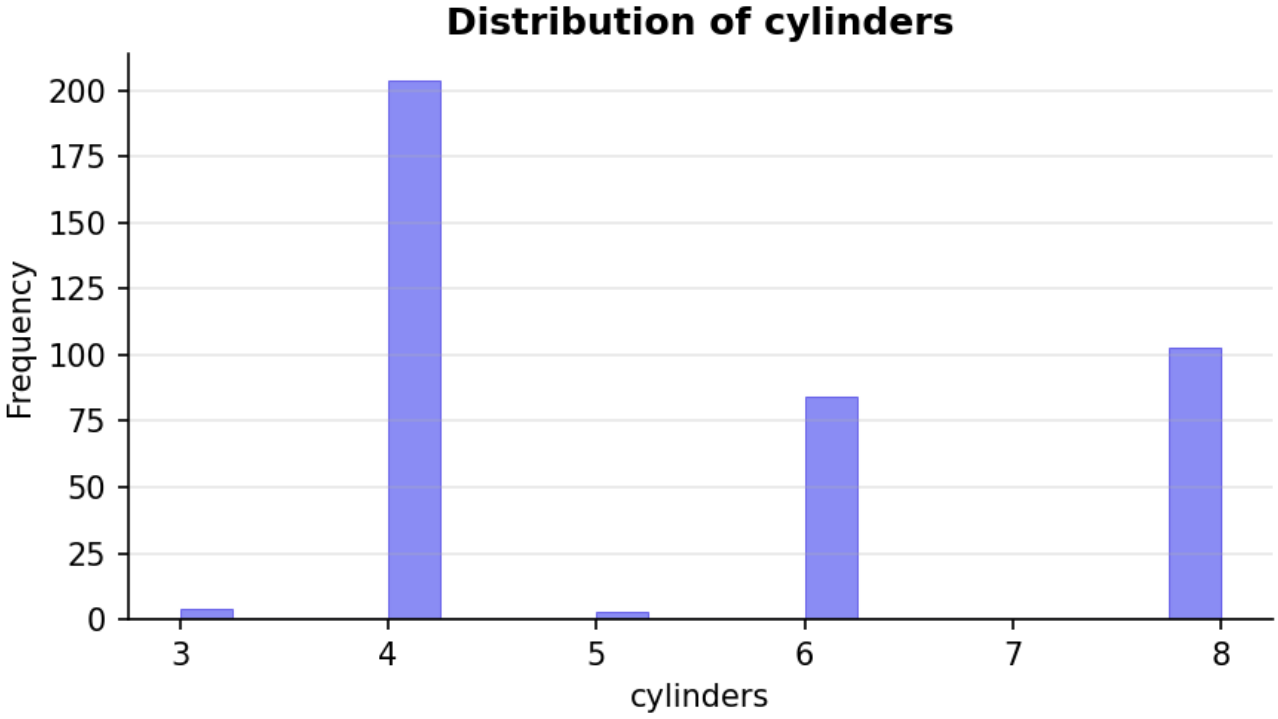
Column	Outliers	%	Q1	Q3	IQR	Lower Bound	Upper Bound

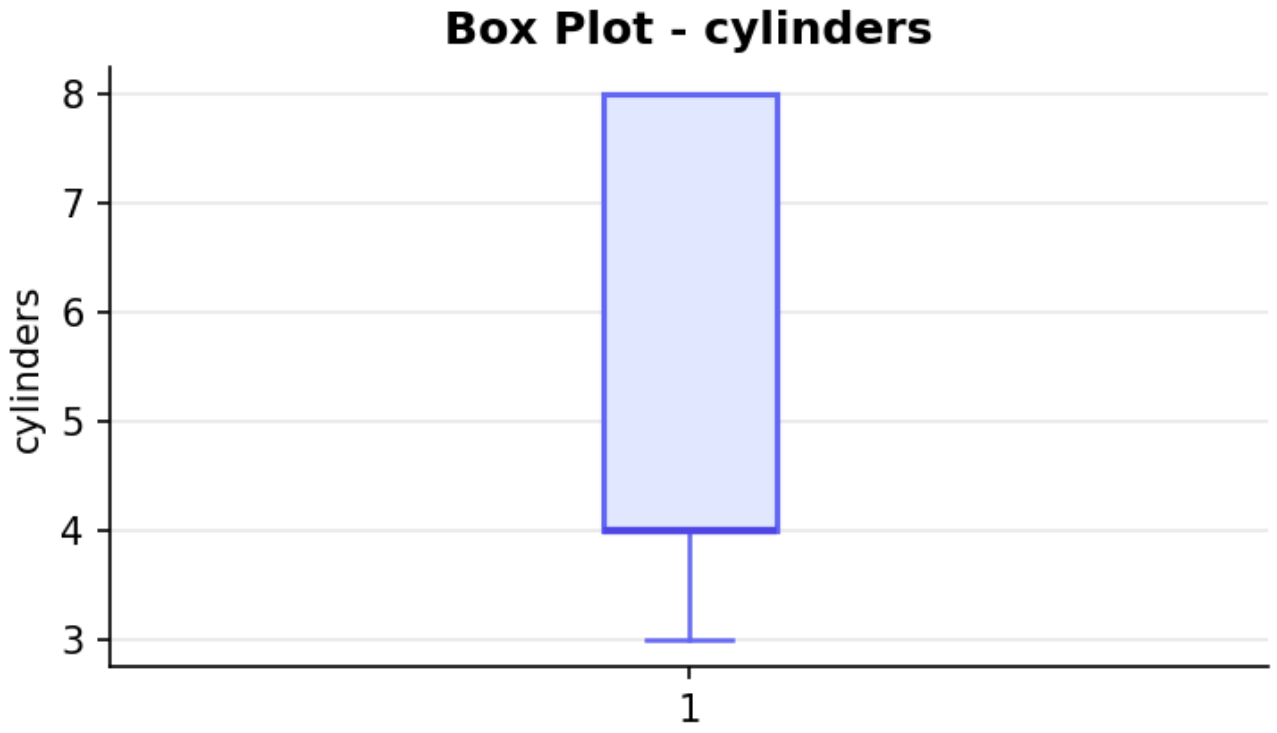
mpg	1	0.3%	17.50	29.00	11.50	0.25	46.25
cylinders	0	0.0%	4.00	8.00	4.00	-2.00	14.00
displacement	0	0.0%	104.25	262.00	157.75	-132.38	498.62
weight	0	0.0%	2223.75	3608.00	1384.25	147.38	5684.38
acceleration	7	1.8%	13.83	17.18	3.35	8.80	22.20
model year	0	0.0%	73.00	79.00	6.00	64.00	88.00
origin	0	0.0%	1.00	2.00	1.00	-0.50	3.50

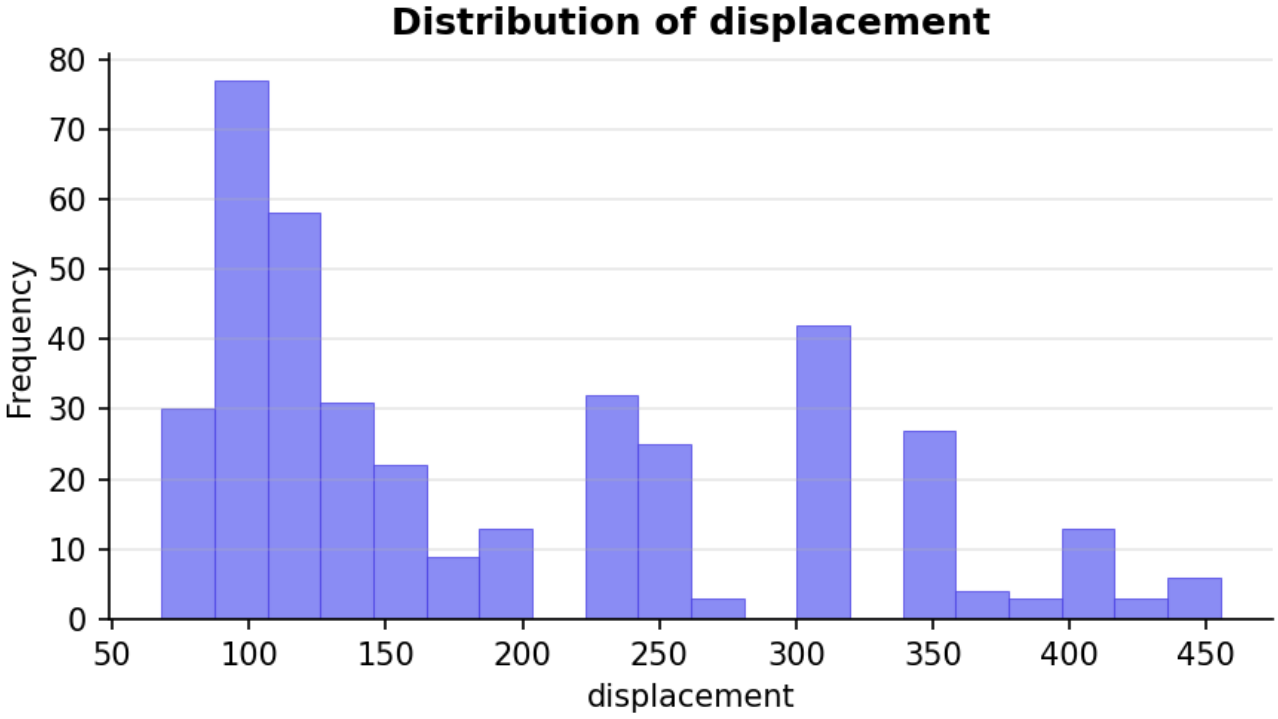
Distribution Analysis

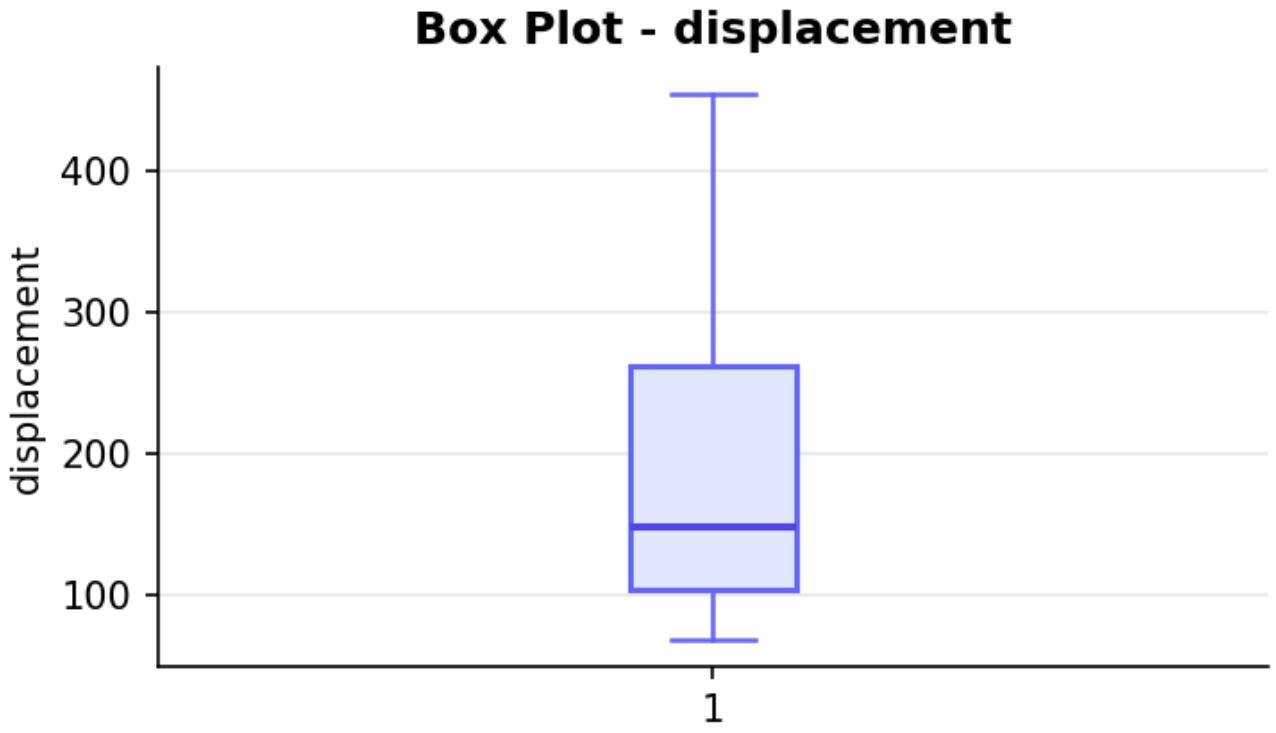


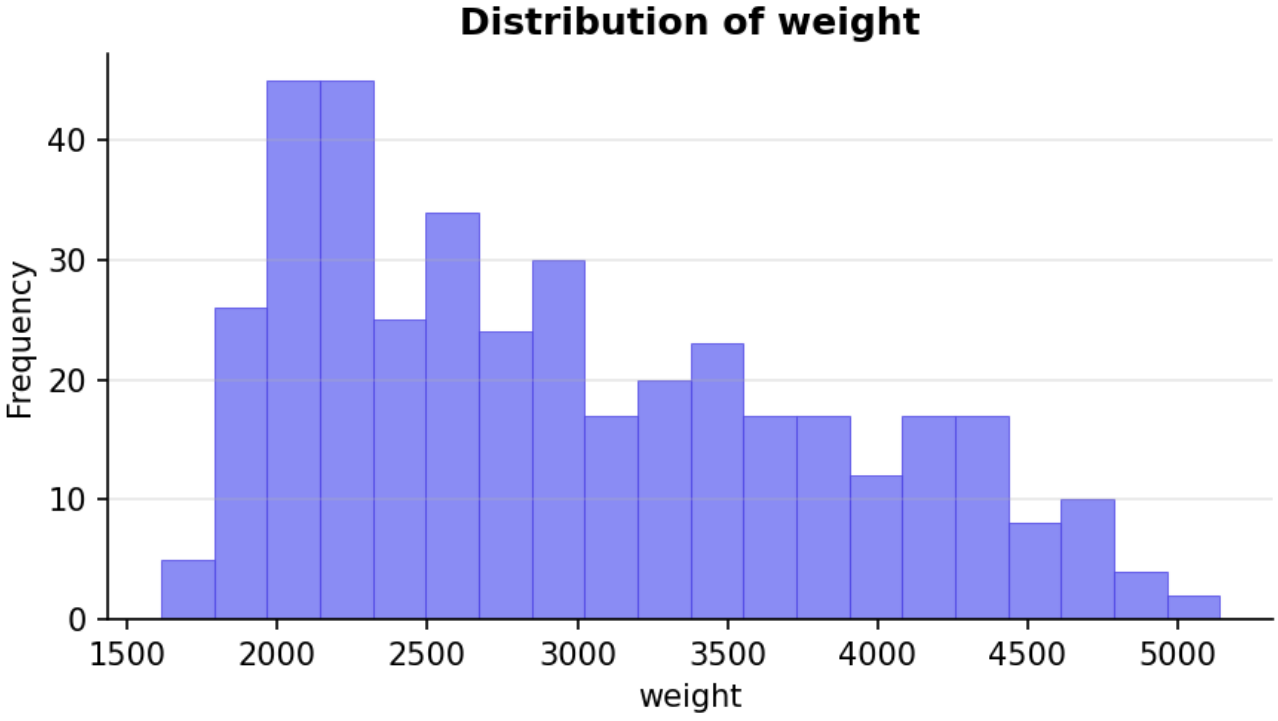

Box Plot - mpg
mpg
40
30
20
10
1

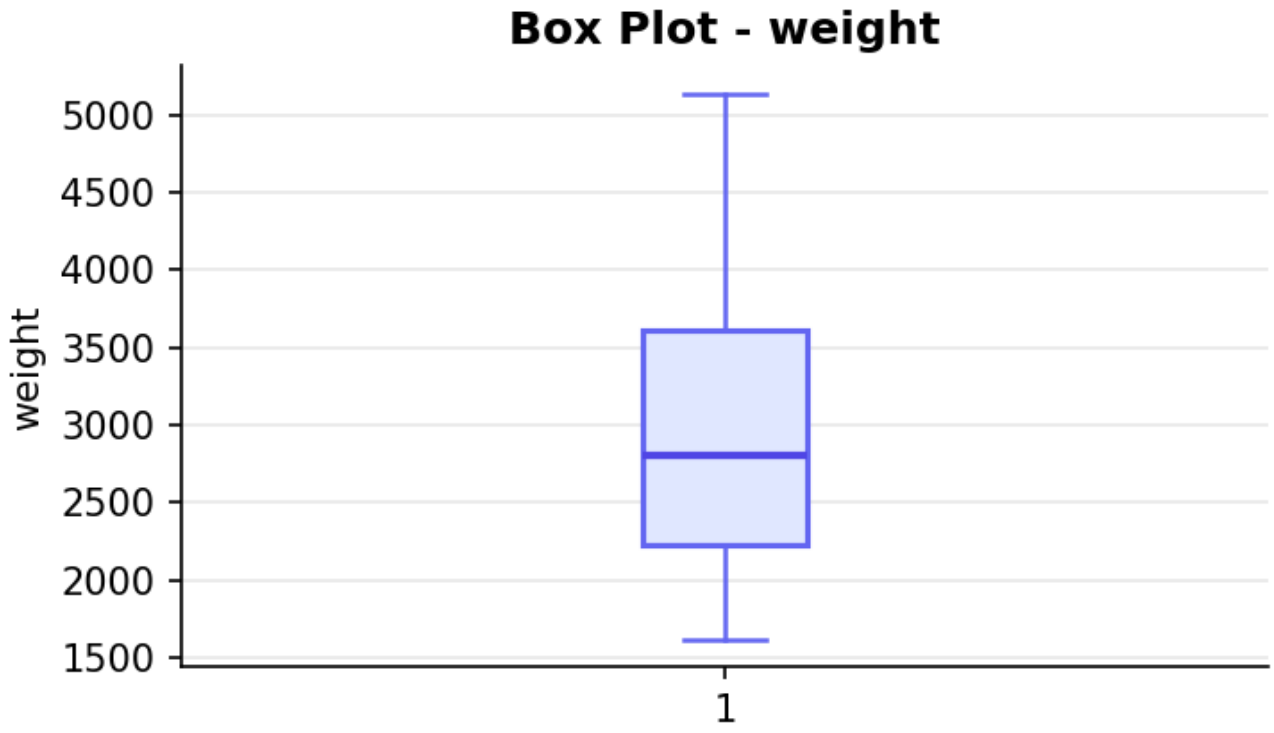

Distribution of cylinders
Frequency
200
175
150
125
100
75
50
25
0
3
4
5
6
7
8
cylinders

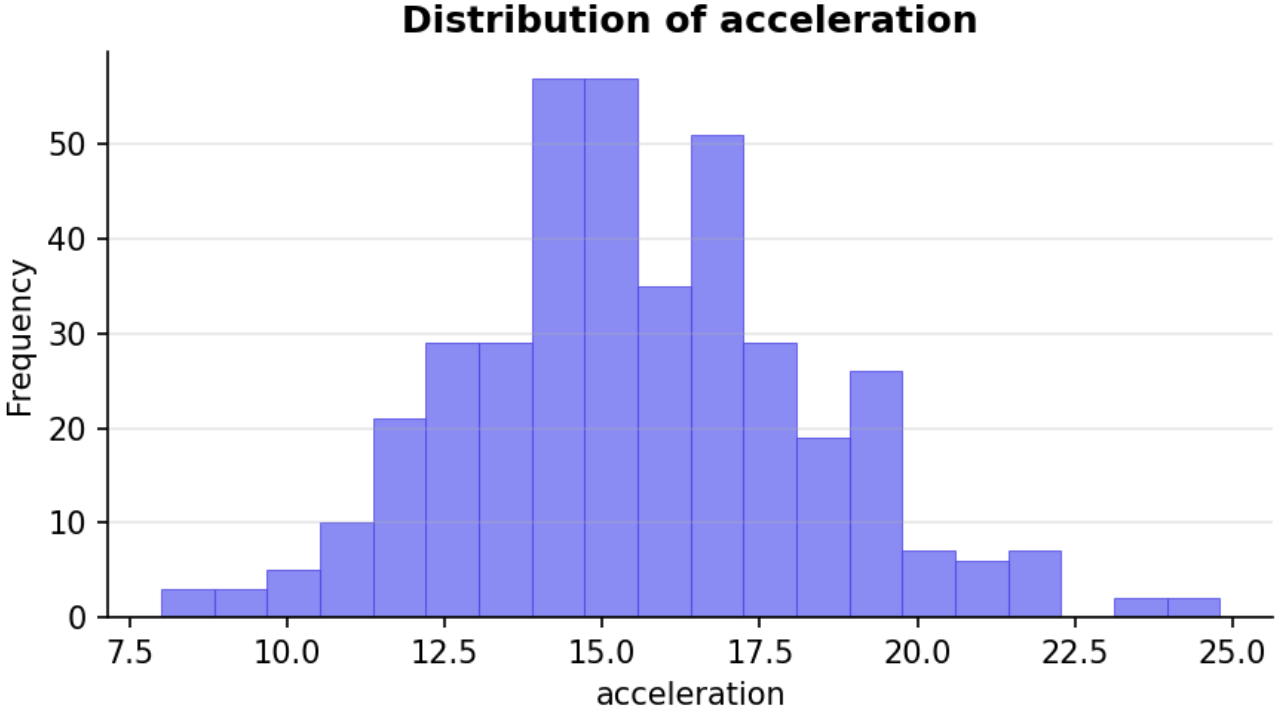

Box Plot - cylinders
cylinders
8
7
6
5
4
3
1

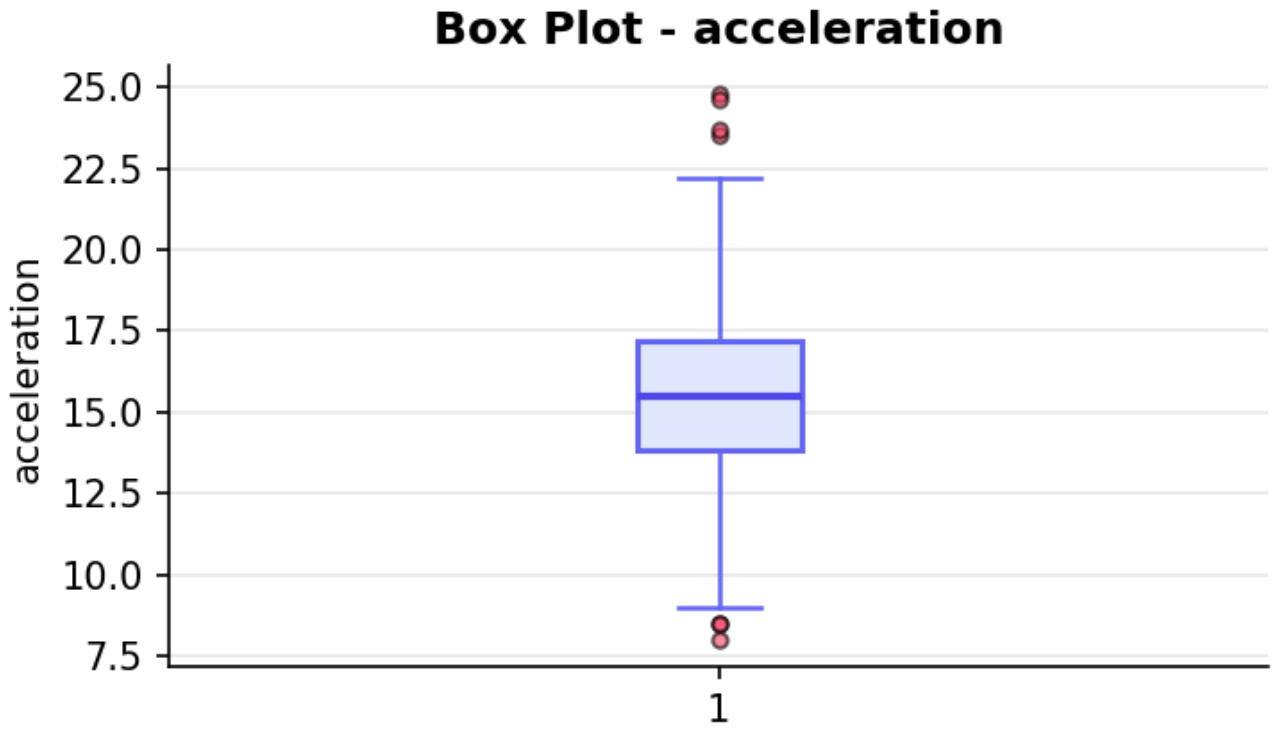

Distribution of displacement
Frequency
80
70
60
50
40
30
20
10
0
50
100
150
200
250
300
350
400
450
displacement

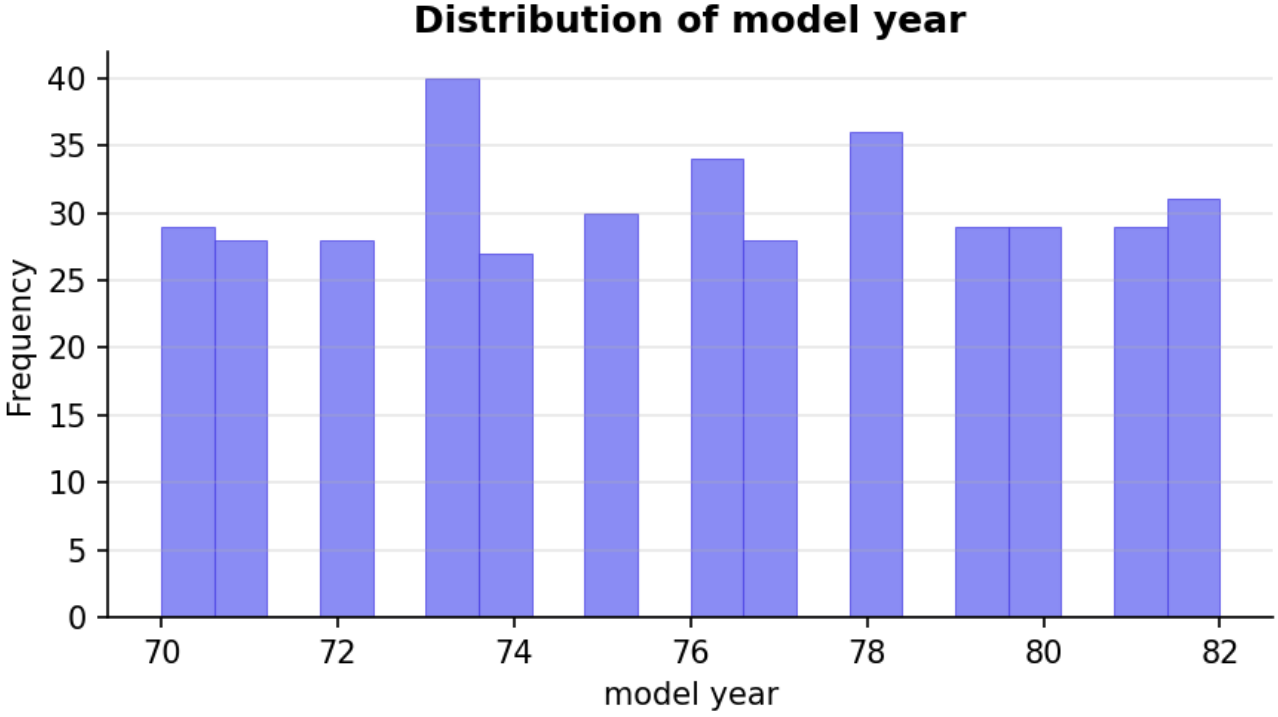

Box Plot - displacement
displacement
400
300
200
100
1

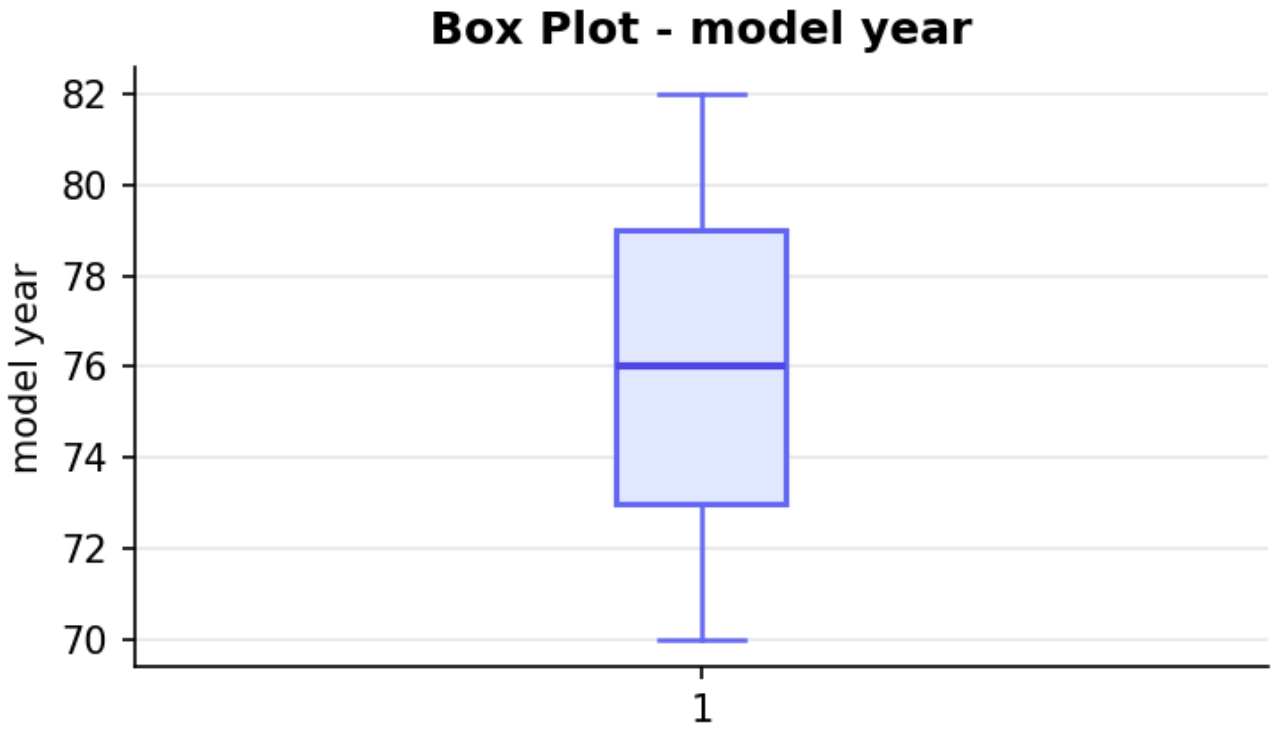

Distribution of weight
Frequency
40
30
20
10
0
1500
2000
2500
3000
3500
4000
4500
5000
weight

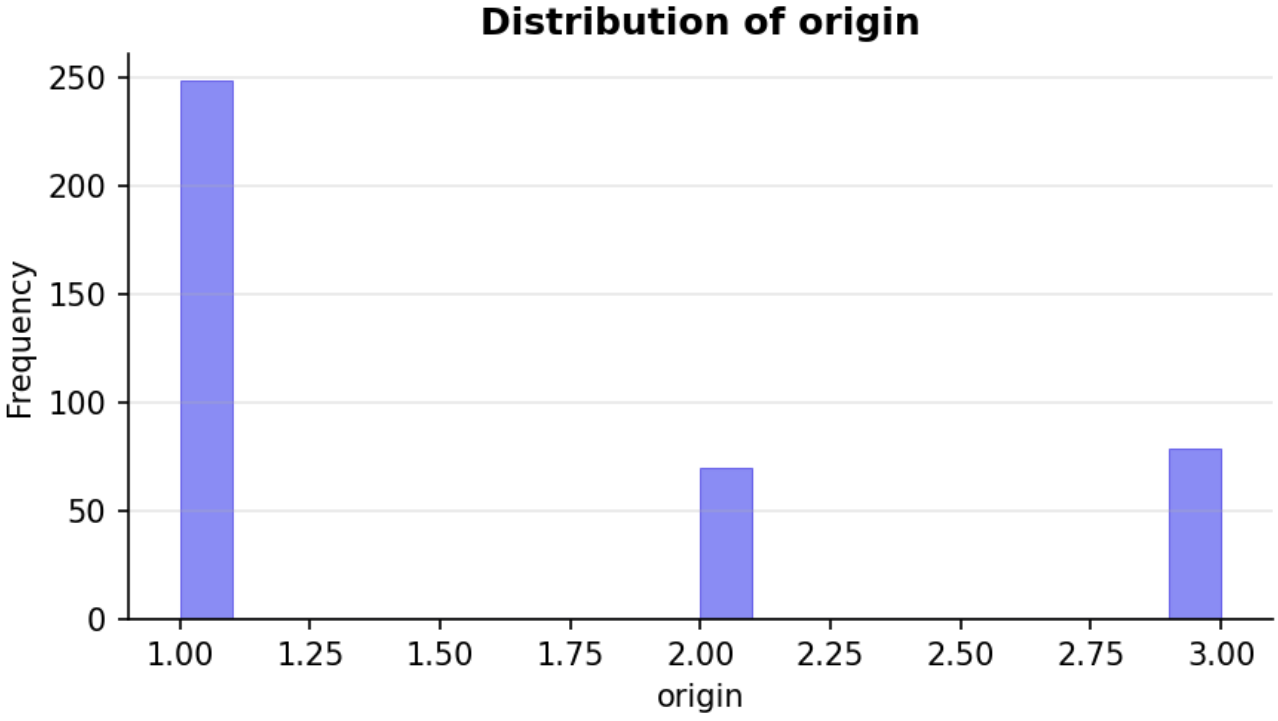

Box Plot - weight
weight
5000
4500
4000
3500
3000
2500
2000
1500
1

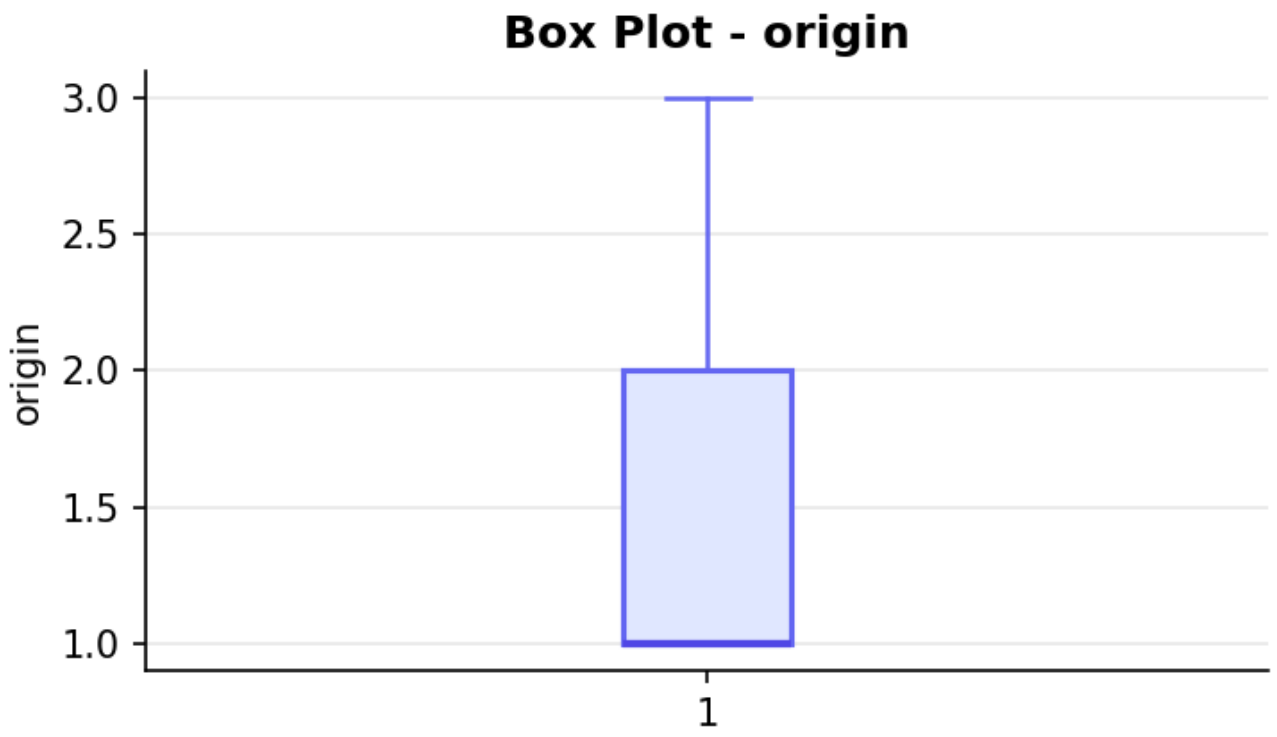

Distribution of acceleration
Frequency
50
40
30
20
10
0
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0
acceleration


Box Plot - acceleration
acceleration
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
1


Distribution of model year
Frequency
40
35
30
25
20
15
10
5
0
70
72
74
76
78
80
82
model year


Box Plot - model year
model year
82
80
78
76
74
72
70
1


Distribution of origin
Frequency
250
200
150
100
50
0
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
origin


Box Plot - origin
origin
3.0
2.5
2.0
1.5
1.0
1